



सूचना का
अधिकार



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Network Lab
नेटवर्क लैब

Experiment Number 1
प्रयोग संख्या 1

Verification of Thevenin's
Theorem

थेवेनिन प्रमेय का सत्यापन

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY,
BHOPAL

NETWORK LAB
EXPERIMENT NO.1

Aim: To study & verify Thevenin's theorem.

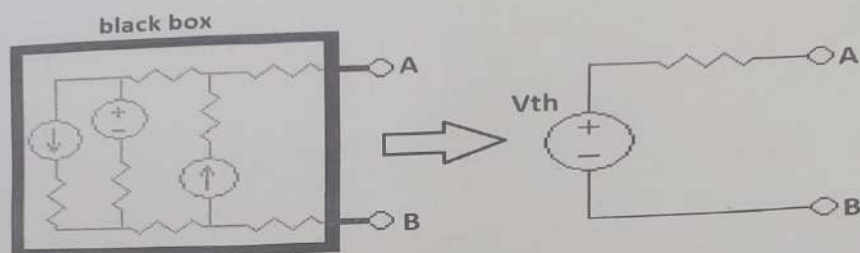
Apparatus Required:

1. Trainer kit
2. Connecting cords

Theory:

Theorem: First of all, Thevenin's theorem states that:

- Any linear electrical network with voltage and current sources and only resistances can be replaced at terminals A-B by an equivalent voltage source V_{th} in series connection with an equivalent resistance R_{th} .
- This equivalent voltage V_{th} is the voltage obtained at terminals A-B of the network with terminals A-B open circuited.
- This equivalent resistance R_{th} is the resistance obtained at terminals A-B of the network with all its independent current sources open circuited and all its independent voltage sources short circuited.



Any black box containing resistances only and voltage and current sources can be replaced to a Thevenin's equivalent circuit consisting of an equivalent voltage source in series connection with an equivalent resistance.

We can verify maximum power transfer theorem with the help of Thevenin's theorem.

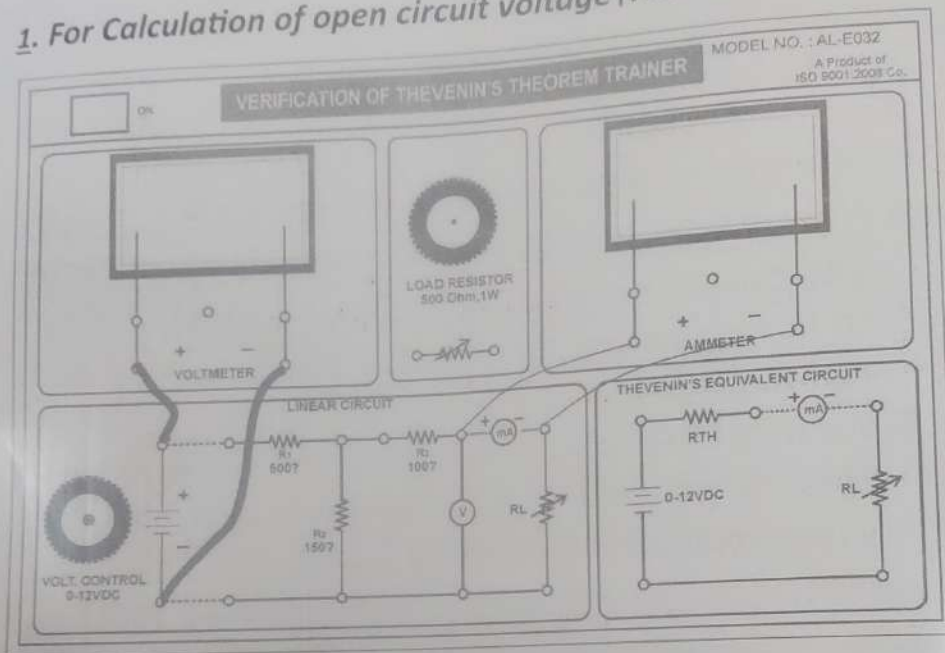
The power transferred from a supply source to a load is at its maximum when the resistance of the load is equal to the internal resistance of the source. On the other words" A resistive load will be consumptive maximum power from the supply when the load resistor is equal to the equivalent (Thevenin) network resistor"

Procedure:

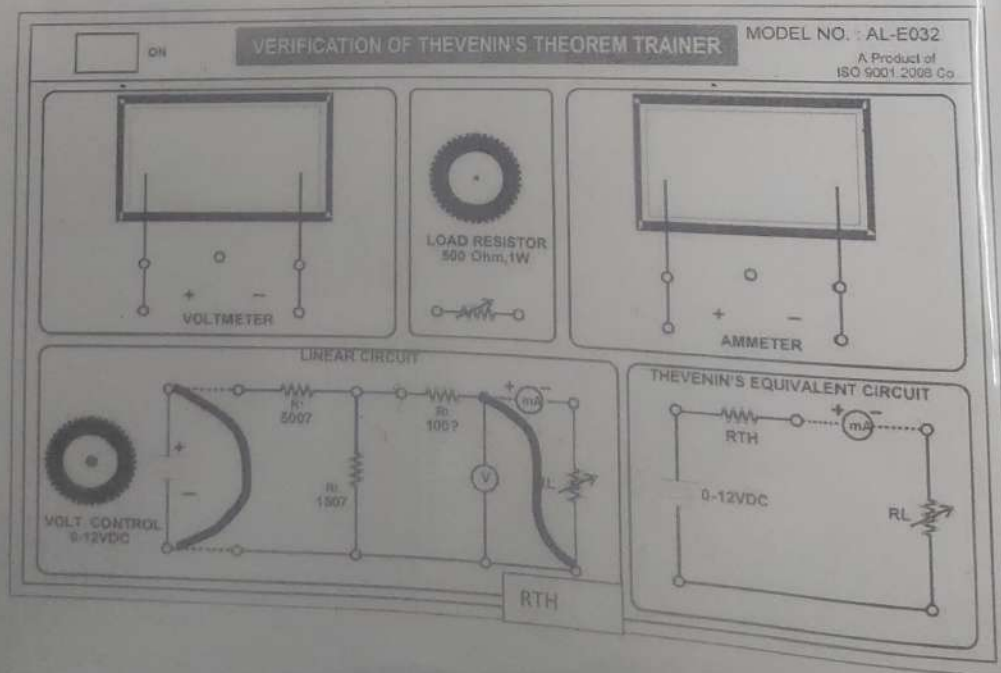
1. Connections should be as per the circuit diagram.
2. Short circuit the voltage source and open circuit the output terminal for calculation of equivalent resistance.
3. Connect the supply voltage and open the output terminal and we get Thevenin's voltage (V_{th}).
4. Now make a equivalent circuit with load connected and calculate the voltage across load (R_L).
5. Using formula find out the Thevenin's voltage (V_{th}) and equivalent resistance (R_{eq}).
6. see here by the current at the resistance equal to the Thevenin's equivalent resistance have maximum, other will be reduced and increased depends upon the resistance.
7. Verify that power is maximum when $R_L = R_{eq}$.

Connection diagram:

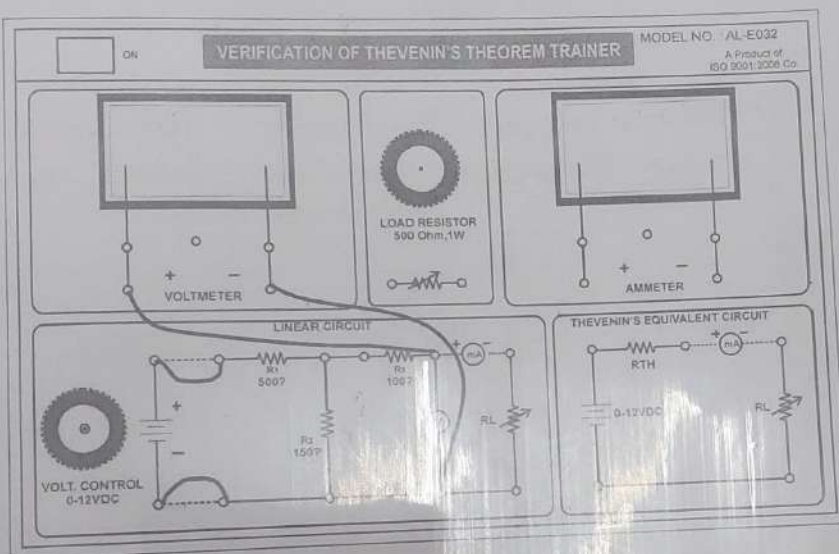
1. For Calculation of open circuit voltage (Thevenin's voltage)



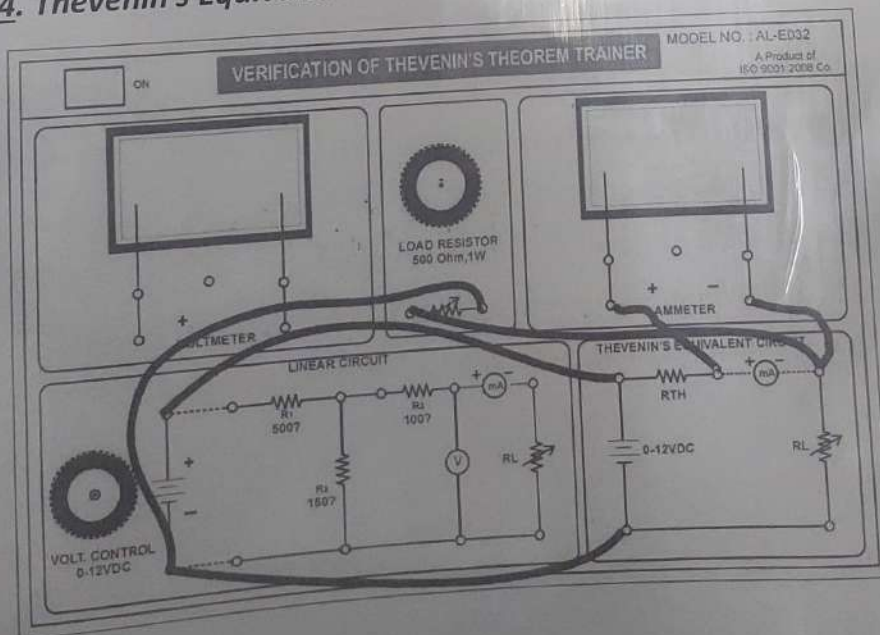
2. For calculation of equivalent resistance (R_{TH}) by Multi meter



3. Calculation of V_{TH}



4. Thevenin's Equivalent circuit with load



Sr.
No.

Observations and calculations:

Sr. no.	Vin	Vth (Thevenin voltage)		Req (equivalent Theoretical resistance)	RL (Load resistance)	
		Practical	Theoretical		Practical	Theoretical
1.						
2.						
3.						
4.						
5.						

Result: If Practical and theoretical values meet near approximation. Then required result is verified.

Questions :

- Q.1 What is Thevenin's Theorem?
- Q.2 What is Thevenin Voltage?
- Q.3 Where is Thevenin's Theorem used?
- Q.4 Is Thevenin's Theorem applicable to non-linear circuits?
- Q.5 Is Thevenin's Theorem applicable to AC circuits?

मौला

उद्देश्य: थेवेनिन

आवश्यक उपकरण

1. ट्रेनर किट

2. कनेक्टिंग का

लिखित:

प्रमेय: सबसे प

• वोल्टेज और

नेटवर्क को ट

प्रतिरोध Rth

• यह समतुल्य

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सभी स्वतंत्र

सर्किट हैं।